

SYNOPSIS

On 3 December 2015, the Panama registered pure car carrier *City of Rotterdam* collided with the Danish registered ro-ro ferry *Primula Seaways* on the River Humber, UK. Both vessels were damaged but made their way to Immingham without assistance. There was no pollution and there were no serious injuries.

The MAIB investigation identified that the outbound *City of Rotterdam* had been set to the northern side of the navigable channel and into the path of the inbound ferry, but this had not been corrected because the pilot on board had become disoriented after looking through an off-axis window on the semi-circular shaped bridge. The car carrier was of an unconventional design and his disorientation was due to 'relative motion illusion', which caused the pilot to think that the vessel was travelling in the direction in which he was looking. Consequently, the pilot's actions, which were designed to manoeuvre the car carrier towards the south side of the channel, were ineffective.

That the pilot's error was allowed to escalate the developing close quarters situation to the point of collision was due to: intervention by *City of Rotterdam*'s master was too late, and the challenges to the pilot's actions by *Primula Seaways*' bridge team and the Humber Vessel Traffic Service being insufficiently robust. Although *Primula Seaways* started to reduce speed about 2 minutes before the collision, a more substantial reduction in speed was warranted.

Following the accident, and an early MAIB recommendation, action has been taken by Fairmont Shipping (Canada) Limited, *City of Rotterdam*'s managers, to reduce the likelihood of relative motion illusion and to improve the bridge resource management of its deck officers. Action has also been taken by Associated British Ports, the harbour authority for the River Humber, to confirm the competency of the pilot and the suitability of *Primula Seaways*' master to hold a pilotage exemption certificate.

Bureau Veritas, *City of Rotterdam*'s classification society, has been recommended to propose measures to the International Association of Classification Societies that are aimed at raising the awareness of relative motion illusion and promoting the need for naval architects and shipbuilders to adhere to internationally accepted ergonomic principles for bridge design.

SECTION 1 - FACTUAL INFORMATION

1.1 PARTICULARS OF *CITY OF ROTTERDAM*, *PRIMULA SEAWAYS* AND ACCIDENT

SHIP PARTICULARS		
Vessel's name	<i>City of Rotterdam</i>	<i>Primula Seaways</i>
Flag	Panama	Denmark (Int.Register)
Classification society	Bureau Veritas	Lloyd's Register
IMO number/fishing numbers	9473468	9259513
Type	Pure car carrier	Ro-ro freight ferry
Registered owner	Picer Marine S.A.	DFDS Seaways AB
Manager(s)	Fairmont Shipping (Canada) Ltd.	DFDS
Construction	Steel	Steel
Year of build	2011	2004
Length overall	139.99m	200.00m
Gross tonnage	21143	32289
Minimum safe manning	14	11
Authorised cargo	Vehicles	Ro-ro freight
Draught (Max)	6.3m	7.2m
VOYAGE PARTICULARS		
Port of departure	Immingham, UK	Gothenburg, Sweden
Port of arrival	Newcastle, UK	Immingham, UK
Type of voyage	Coastal	International
Cargo information	In ballast	Ro-ro freight
Manning	18	19
MARINE CASUALTY INFORMATION		
Date and time	3 December 2015 at 2040 UTC	
Type of marine casualty or incident	Serious Marine Casualty	
Location of incident	River Humber, UK 53°35.1N, 000°02.6E	
Place on board	Bow/forecastle	Bow/forecastle
Injuries/fatalities	None	None
Damage/environmental impact	Damage to bow, forecastle, port shoulder, port bilge keel and ballast tank.	Damage to bow and forecastle.
Ship operation	On passage	On passage
Voyage segment	Transit	Transit
External & internal environment	Wind: south-south-west gusting to 40kts. It was dark with clear skies. The visibility was good and the tidal stream was flooding at about 1.5kts	
Persons on board	18 crew	19 crew 6 passengers

1.2 NARRATIVE

1.2.1 Events leading up to the collision

On 3 December 2015, the Panama registered pure car carrier *City of Rotterdam* was preparing to depart Immingham Dock, UK (**Figure 1**). On the bridge were the master, the third officer, an able seaman (AB) helmsman and a pilot. The master and the pilot discussed the departure with reference to the vessel's pilot information card and the pilot's passage plan.

During the exchange, the master advised the pilot to stand at either the forward centreline conning station or behind the navigation workstation (**Figure 2**). He also pointed out to the pilot a length of cord on the centre window that indicated the vessel's centreline. The master and the pilot also discussed the potential effects of the wind. The vessel was high-sided and in ballast, and the wind was gusting up to 40 knots from the south-south-west. The pilot anticipated that the vessel would experience a high drift rate throughout its passage, but more so once passed Grimsby where it was more exposed and the effect of the tidal stream would be more pronounced.

At 1858, *City of Rotterdam* sailed from its berth and the pilot conned the vessel in to Immingham lock. By 1959, the vessel had cleared the lock, released the attending tugs and was passing 9A buoy in the main navigation channel in the River Humber (**Figure 3**). The vessel was on a south-easterly heading in manual steering at a

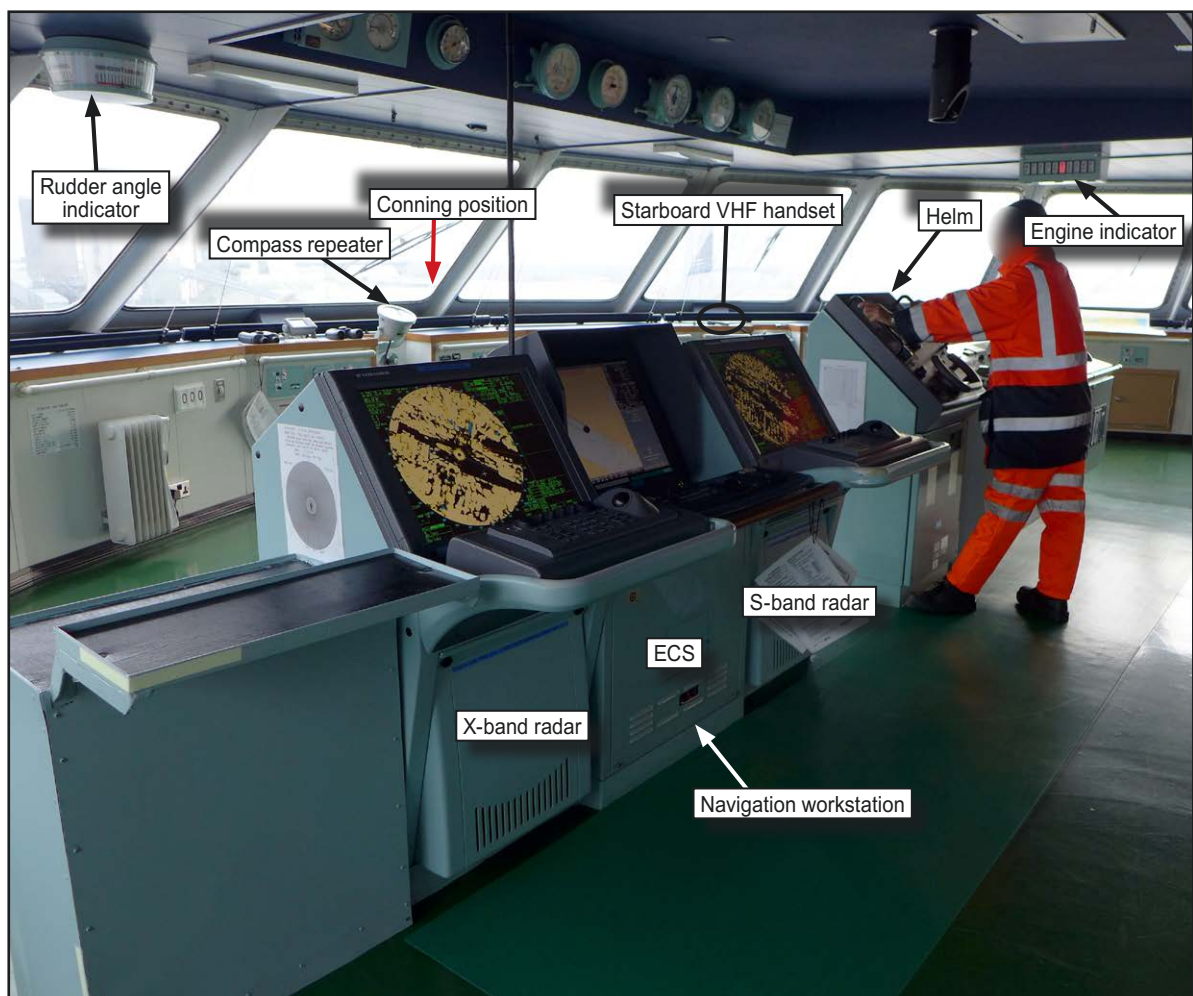


Figure 2: *City of Rotterdam* - bridge (from port side)

speed of 12kts¹. The tidal stream was flooding at a rate of about 1.5kts. On passing 9A buoy, the pilot reported the vessel's position to Humber Vessel Traffic Service (VTS) via very high frequency (VHF) radio channel 12². As *City of Rotterdam* continued on passage down the main navigation channel, the pilot primarily monitored the vessel's position by eye but he also used the electronic chart system (ECS) and the port radar display (**Figure 2**) (the starboard radar display was on standby).

At 2027, *City of Rotterdam* passed the Grimsby Middle buoy on a heading of 125° (**Figure 4**). At that time, the Denmark registered ro-ro freight ferry *Primula Seaways* was in the outer approaches of the River Humber, inbound for Immingham. The ferry was on a trackpilot controlled heading of 291° at 20kts and was overtaking the Malta registered bulk carrier *Seferis*. On the ferry's bridge were the master, second officer and an AB. The second officer was seated in the chair to starboard of the central conning station (**Figure 5**). He was monitoring the ferry's position on the Electronic Chart Display and Information System (ECDIS) and by radar. The master had the control of navigation and was seated in the chair on the central console's port side.

Between 2027 and 2032, *City of Rotterdam*'s pilot altered the vessel's heading from 125° to 095° in 5° increments (**Figure 6**). The vessel was to the north of the intended track and the pilot informed the master that he would try and manoeuvre the ship further to the south. By that time, *Primula Seaways* was in clear sight and the pilot informed the master that the vessels would pass port to port.

Meanwhile, the VTS operator monitoring VHF channel 12 had reported *City of Rotterdam*'s northerly position to the watch manager. However, the watch manager was not concerned as the pilot had sufficient time to take corrective action.

At 2034, *Primula Seaways*' master called VTS on VHF channel 12 (**Table 1**).

<i>Primula Seaways</i> (master)	VTS Humber <i>Primula Seaways</i>
VTS Humber Channel 12 operator	<i>Primula Seaways</i> VTS Humber
<i>Primula Seaways</i> (master)	Yeah, good evening just a question. <i>City of Rotterdam</i> is he going for the Hawke anchorage?
VTS (Channel 12 operator)	<i>Primula Seaways</i> VTS negative. The <i>City of Rotterdam</i> is proceeding outward with pilot.
<i>Primula Seaways</i> (master)	He's proceeding outward [pause] OK.

Table 1: VHF radio exchange between *Primula Seaways* and Humber VTS between 2034 and 2035

City of Rotterdam and *Primula Seaways* were 2.8nm apart. The car carrier was now heading 105° at 12.2kts and *Primula Seaways* was heading 295° at 16.5kts (**Figure 7**). The ferry's speed had been reduced in order to meet its scheduled arrival time. At 2035:44, *City of Rotterdam*'s pilot reported the vessel's position to VTS on VHF

¹ All speeds are speed over the ground

² See paragraphs 1.4.5 and 1.15.4

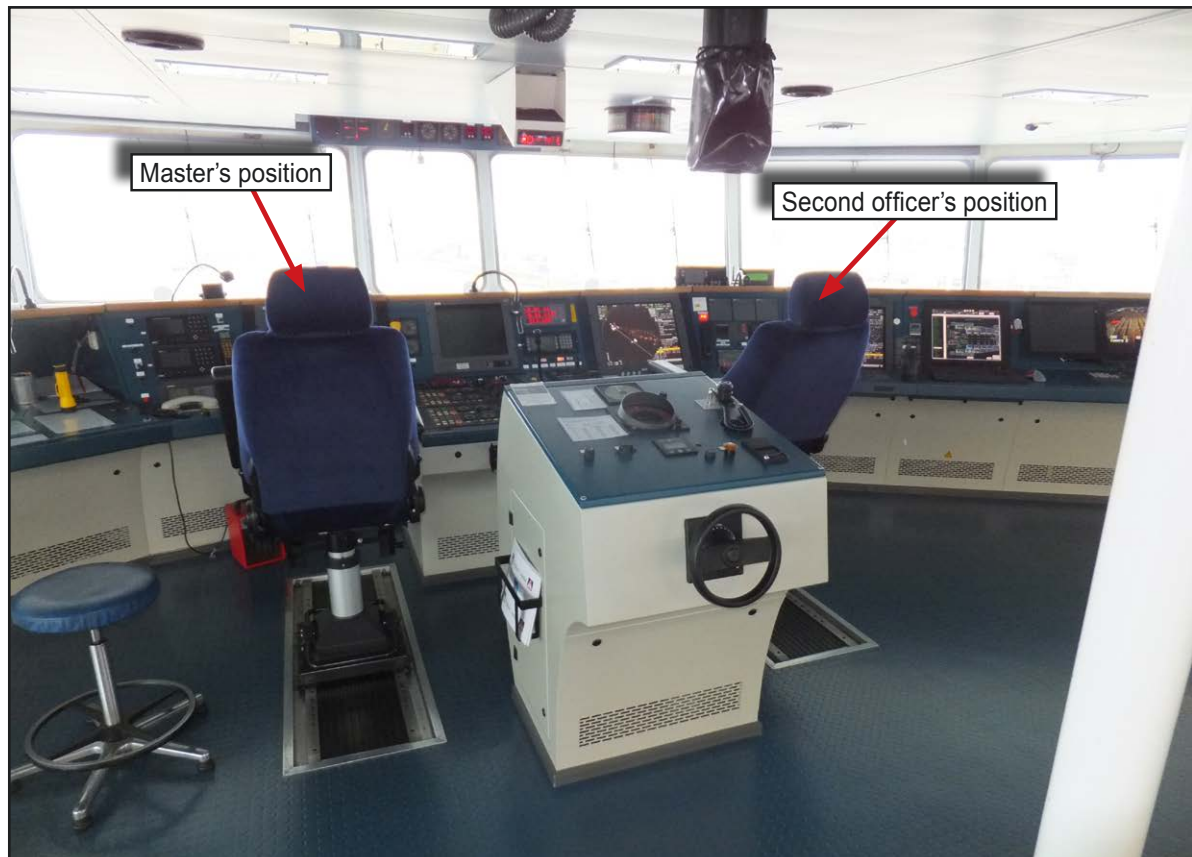


Figure 5: *Primula Seaways* - bridge

channel 14 (**Table 2**) as it approached 4A Clee Ness light float. The pilot used the VHF radio sited at the front of the bridge to starboard of the centreline (**Figure 2**) and had to wait for up to 20 seconds for VTS to complete a VHF exchange with another vessel before he could start his transmission.

Pilot	<i>VTS the City of Rotterdam</i>
VTS (Channel 14 operator)	<i>City of Rotterdam VTS</i>
<i>City of Rotterdam</i> (pilot)	<i>Yeah, good evening sir, I'm Clee Ness out and New Sand Hole³</i>
VTS (Channel 14 operator)	<i>Clee Ness New Sand Hole, thank you City of Rotterdam, tide gauge Spurn Head four decimal six five [pause] launch this evening the Saturn [pause] the Saturn has three boardings to do at alpha whisky two⁴ and then she'll follow you outwards</i>
<i>City of Rotterdam</i> (Pilot)	<i>Roger that, thanks</i>

Table 2: VHF radio exchange between *City of Rotterdam* and Humber VTS between 2035:44 and 2036:17

³ New Sand Hole is a reference to the route to be taken by the vessel once it has passed the Alpha buoy. There are three channels: New Sand Hole (NE), Sea Reach (E) and Ross Reach (SE).

⁴ AW2 is a pilot boarding point used in adverse weather.

Primula Seaways' master and second officer continued to be concerned that *City of Rotterdam* remained on the northern side of the channel and did not appear to be altering course. The VTS watch manager was now also concerned, and he and *Primula Seaways*' second officer called the car carrier in quick succession on VHF channel 14 at 2037 (**Table 3**). The pilot replied, again using the VHF radio on the starboard side of the bridge front.

VTS	<i>City of Rotterdam, VTS</i>
<i>Primula Seaways</i> (second officer)	<i>City of Rotterdam City of Rotterdam, Primula Seaways Primula Seaways</i>
<i>City of Rotterdam</i> (pilot)	<i>Station calling City of Rotterdam thirteen please</i>
<i>Primula Seaways</i> (second officer)	<i>One three</i>
<i>Primula Seaways</i> (second officer)	<i>City of Rotterdam Primula Seaways</i>
<i>City of Rotterdam</i> (pilot)	<i>Loud and clear good evening sir go ahead I'm trying to drop her down to the south as much as possible is that you on my port bow?</i>
<i>Primula Seaways</i> (second officer)	<i>Yeah that's right, we're just coming up to Clee Ness now</i>
<i>City of Rotterdam</i> (pilot)	<i>Yeah I'm trying to bring her as far to the south as the wind will allow me but er [pause] yeah I'll keep coming down to the south more positive</i>
<i>Primula Seaways</i> (second officer)	<i>I'd be obliged for that ok back to fourteen</i>

Table 3: VHF radio exchange between *Primula Seaways* and *City of Rotterdam* between 2037:09 and 2038:02

1.2.2 The collision

At 2038, the distance between *City of Rotterdam* and *Primula Seaways* was 0.97nm. The ferry continued on a heading of 295° (**Figure 8**) and its port and starboard sidelights were visible to *City of Rotterdam*'s bridge team. *City of Rotterdam*'s pilot confirmed with the helmsman that the vessel's heading was now 110°. He then ordered the helmsman to steer 115°.

At 2038:25, the VTS watch manager called *City of Rotterdam* on VHF channel 14 (**Table 4**). *Seferis* initially responded, but the VTS watch manager continued with his message.

VTS (watch manager)	<i>City of Rotterdam City of Rotterdam VTS</i>
<i>Seferis</i>	<i>Seferis replying</i>
VTS (watch manager)	<i>Yeah City of Rotterdam VTS from my equipment you're tracking to the north... tracking into the Hawke anchorage at the moment</i>
<i>City of Rotterdam</i> (pilot)	<i>Yeah we're on full speed and heading nearly south but we're going to try and bring her as far down as possible</i>
VTS (watch manager)	<i>Yeah you're entering the Hawke anchorage now [pilot's name], the Primula Seaways is right to the north of the channel, you might struggle to get south of him now</i>
<i>City of Rotterdam</i> (pilot)	<i>Yeah I'm on full speed and heading right down south</i>

Table 4: VHF radio exchange between VTS Humber and *City of Rotterdam* at 2038:25 and 2039:15

During the VHF exchange, *Primula Seaways'* master reduced the ferry's engine telegraph to 'half ahead', which equated to a speed through the water of 9.4kts.

City of Rotterdam's pilot ordered "starboard 20". As the car carrier's heading reached 125° the pilot ordered "midships" then "135°". Accordingly, the helmsman arrested the vessel's swing to starboard in order to steady the vessel as ordered. The car carrier's master expressed concern over the developing situation and the pilot explained to him that both vessels were experiencing drift.

At 2039:27, *Primula Seaways'* bridge team realised that *City of Rotterdam* was not turning to starboard as quickly as they expected. Manual steering was selected and full starboard helm was applied. The engine was also set to 'full astern'.

At 2039:57, *City of Rotterdam's* pilot ordered "150°" and the helmsman applied 5° of starboard helm. Five seconds later, the master shouted "what is he doing?" The two vessels were now 0.27nm apart. *City of Rotterdam's* speed was 12kts and *Primula Seaways'* speed was 14.3kts.

The VTS watch manager called *Primula Seaways* on VHF channel 14 (**Table 5**).

VTS (watch manager)	<i>Primula Seaways VTS</i>
<i>Primula Seaways</i> (second officer)	<i>VTS Primula Seaways go ahead</i>
VTS (watch manager)	<i>Yes sir, warning, it appears that the City of Rotterdam is unable to bring it down further to the south [pause] from my equipment it appears there is a risk of collision</i>
<i>Primula Seaways</i> (second officer)	<i>Er [unintelligible] we can't do much now we're coming hard a starboard but I think we're going to collide [pause] yep</i>
VTS (watch manager)	<i>City of Rotterdam VTS did you receive?</i>

Table 5: VHF radio exchange between VTS Humber and *Primula Seaways* at 2040

City of Rotterdam did not reply. At 2040:23, the vessel's master shouted “go to starboard”. He then ordered “midships” followed by “hard to port”. Fourteen seconds later, *Primula Seaways* and *City of Rotterdam* collided, port bow to port bow (**Figure 9**), on headings of 288° and 163° respectively.

1.2.3 Post-collision actions

Primula Seaways

Primula Seaways' second officer reported the collision to the VTS while the master turned the ferry back to its intended heading at slow speed. The general alarm was not sounded but the master made a public address system announcement to advise the vessel's crew and six passengers of the situation. The chief engineer and chief officer conducted a damage assessment of the vessel. On completion, the master reported to the VTS that there was “no apparent damage” and the VTS permitted the ferry to continue its passage to Immingham. The information that had been recorded on the vessel's voyage data recorder (VDR) was saved.

City of Rotterdam

On impact, *City of Rotterdam* heeled to starboard. The master and pilot fell to the deck but they were not injured. The pilot reported the collision to the VTS and the car carrier was manoeuvred towards the Bull anchorage at slow speed. The general alarm was not sounded. The chief and second officers went to the bridge immediately. The chief officer then went forward to assess the damage while the on-watch engineers checked for damage in the machinery spaces.

City of Rotterdam's forward hydraulic system was badly damaged so the vessel was unable to anchor. As soon as it was confirmed that there had been no breach in the car carrier's watertight integrity, the VTS authorised the vessel to return to Immingham. The VDR data was saved, but no radar information was recorded due to the starboard radar display being on standby.

- When *Primula Seaways* started to reduce speed to 'half ahead' at 2038:25, *City of Rotterdam* was within 1nm (**Figure 18**). The car carrier's pilot had informed the ferry's master several times that he was manoeuvring to the south, yet there was no sign of the vessel doing so. From the simulations, it was evident that a course alteration to starboard would have caused the ferry to cross *City of Rotterdam*'s bow while an alteration to port would have exacerbated the close quarters situation that was developing if the car carrier did turn more to the south as the pilot had indicated.



Figure 18: Simulated night-time view from *Primula Seaways* at 2038:25, *City of Rotterdam* is showing its two masthead lights and green sidelight

Figures 19 and 20 show the respective simulated views from *City of Rotterdam* and *Primula Seaways* at 2039:27 when the ferry's engine was put to 'full astern' and full starboard helm was applied.

1.12 SIMILAR BRIDGE DESIGNS

A number of modern vessels are constructed with bridges that are semi-circular, with off-axis windows. In all such vessels identified, the bridges were fitted with integrated navigation systems and fixed seats at the conning positions (**Figure 21**). It is noted that *NS Savannah*, which was launched in 1959, had off-axis windows and, against the convention of the time, the bridge's main workstation was located on the centreline at the bridge front (**Figure 22**). In 2015, the Kyokuyo Shipyard launched the container ship *Natori* (**Figure 23**), another ship with a hemispherical bow. However, on the direction of the vessel's Flag administration (Japan), the bridge was fitted with sloping windows that were compliant with SOLAS V/22.1.9.1.

1.15 RIVER HUMBER

1.15.1 Governance

The River Humber is the busiest ports complex in the UK, with about 30,000 ship movements annually. Associated British Ports (ABP) is the statutory and competent harbour authority¹⁸, but the harbour authority's operational roles are delegated to ABP Humber Estuary Services (HES). ABP operates 22 UK ports and complies with the Port Marine Safety Code¹⁹.

1.15.2 The Bull Channel

The Bull Channel (**Figure 1**) was at its narrowest between the South Shoal Buoy and the 4A Clee Ness buoy, when it was about 4 cables wide. There was no speed limit in the channel but the Humber Byelaws included:

Nothing in these Byelaws shall affect the operation of the Collision Regulations or the duty upon the master of a vessel to comply therewith.

and

The master of a vessel shall navigate the vessel with due care and caution and at a speed and in a manner which shall not endanger the safety of any person or any other vessel or cause damage thereto or to a floating navigational mark or mooring or other property.

The North Sea (West) Pilot²⁰ advises mariners:

the flood stream sets strongly to the north in the Bull Channel (53°35'N 0°03'E) which can carry vessels out of the fairway. It is reported (1994) that vessels have frequently collided with No.5 Gate Buoy which marks the N side of the channel.

1.15.3 Pilotage

Pilotage was compulsory in the Humber for all vessels 60m or over in length. ABP employed approximately 120 pilots, who were authorised following their successful completion of training and oral and practical examinations. The pilots were re-assessed at intervals not exceeding 5 years.

Masters and chief officers of vessels navigating the Humber pilotage area could apply for a PEC. Applicants had to satisfy the harbour authority that they had a suitable level of skill, experience and local knowledge to navigate safely in the harbour area. PECs were required to be re-validated every 12 months (based on achieving the required minimum number of trips), and PEC holders were re-assessed every 5 years.

¹⁸ A competent harbour authority in the UK has statutory powers relating to the provision of pilotage in its waters.

¹⁹ The Port Marine Safety Code is the UK standard for every aspect of port marine safety. It applies to all harbour authorities in the UK that have statutory powers and duties. Continued compliance with the Code is required to be confirmed to the MCA every 3 years.

²⁰ Admiralty Sailing Directions North Sea (West) Pilot NP54.

1.15.4 VTS Humber

VTS Humber provided a traffic organisation service (TOS)²¹ east of the Humber Bridge, which included the area between Immingham and the eastern port limit. It operated to international standards detailed in IMO Resolution A857(20) and the IALA VTS Manual. All of the VTS operators were familiar with and regularly used message markers²² during VHF radio communications.

The VTS station was located on Spurn Head (**Figure 1**) and was equipped with VHF radio, a bespoke VTS chart display system with radar and AIS overlays, land-based closed circuit television coverage at critical points and weather monitoring equipment. Operators were also able to access ABP's port and vessel information system to assist with the planning and monitoring of all significant traffic in the area.

The VTS station operated continuously with three duty operators, a VHF channel 12 operator (west of the 4A Clee Ness Buoy), a VHF channel 14 operator (east of the 4A Clee Ness Buoy) and a watch manager who operated VHF channel 15 (upper Humber). The VHF channel 12 operator was generally the least experienced and the watch manager the most experienced. The VTS operators followed an 8 hours on/8 hours off routine for a period of 4 or 6 days. They lived on site as access to the station was generally only achievable by boat.

1.15.5 Previous accidents

The MAIB has investigated three collisions in the Humber since 2005:

- In January 2005, the oil products tanker *Amenity* and a ro-ro cargo vessel *Tor Dania* collided between the Grimsby Middle and South Shoal buoys. Among the safety issues identified in the MAIB investigation report (MAIB 20/2005) were the training and assessment of PEC holders and bridge manning.
- In June 2006, the general cargo ship *Skagern* and container vessel *Samskip Courier* collided. Both vessels were under pilotage in restricted visibility near the Saltend jetties east of Hull docks. The investigation report (MAIB 6/2007) identified that the masters of both vessels had been over-reliant on the pilots and that the interaction and communication among the bridge teams was poor. Following the accident, the MAIB recommended:

The Port Marine Safety Code Steering Group, to:

- | | |
|----------|--|
| 2007/122 | Promulgate to Port Authorities the need for pilots to maintain dialogue with the bridge team regarding the conduct and execution of the passage plan, thus ensuring the team is kept fully involved, and informed, at all times. |
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²¹ See Paragraph 1.16.

²² The IALA VTS manual lists eight message markers (Instruction, Advice, Warning, Information, Question, Answer, Request and Intention) which can be used at the start of VHF communications to increase the probability of a message being understood.

The International Chamber of Shipping, to:

2007/125 Through its member organisations, emphasise the need for shipowners to ensure masters are given clear guidelines that detail the importance of effective dialogue with pilots, and identifies the need for masters to challenge or question decisions or actions taken by pilots at an early stage so that, when required, effective corrective action can be taken to prevent accidents.

- In April 2007, the products tanker *Audacity* and general cargo ship *Leonis* collided in the precautionary area east of Spurn Head. Among the contributing factors identified in the MAIB investigation report (MAIB 2/2008) were that the pilots and bridge teams did not make a full assessment of the risk of collision and that VTS procedures for managing traffic in the precautionary area were insufficient. As a result of the accident, ABP HES took several actions to, among other things, improve the performance of the VTS and its pilots.

1.16 VESSEL TRAFFIC SERVICES

Three levels of VTS are available: an information service (INS), a traffic organisation service (TOS), and a navigation assistance service (NAS). Authorities can operate one, two or three services from the same VTS station.

The type of service provided by VTS in the UK is detailed in Marine Guidance Note (MGN) 401 (M+F) - *Navigation: Vessel Traffic Services (VTS) and Local Port Services (LPS) in the United Kingdom*. With reference to a TOS, the MGN states:

This service type provides essential and timely information to assist the onboard decision-making process. It may involve the provision of information, advice and instructions. TOS concerns the forward planning of movements to maintain vessel safety and to achieve efficiency. This service may involve:

- *The allocation of water space;*
- *The mandatory reporting of movements;*
- *The position, identity, intention and destination of vessels;*
- *Specific information, such as traffic congestion and advice about vessels with VTS sailing / route plans;*
- *Information such as meteorological and hydrological conditions, notices to mariners, status of aids to navigation;*
- *Amendments and changes in promulgated information concerning the VTS area such as boundaries, procedures, radio frequencies, reporting points;*
- *Establishing routes to be followed and speed limits to be observed and such other measures as may be considered necessary and appropriate by the VTS;*
- *Establishing and operating a system of traffic clearances - all or certain classes of vessels may be required to participate in this service and shall not proceed without clearance;*
- *Specific information such as traffic congestion and special vessels with limited manoeuvrability which may impose restrictions on the navigation of other vessels or any other potential hindrances.*

SECTION 2 - ANALYSIS

2.1 AIM

The purpose of the analysis is to determine the contributory causes and circumstances of the accident as a basis for making recommendations to prevent similar accidents occurring in the future.

2.2 THE COLLISION

The collision between *Primula Seaways* and *City of Rotterdam* stemmed from the latter being set to the northern side of the Bull Channel by the wind and the tidal stream, followed by the distortion of its pilot's spatial awareness due to a 'relative motion illusion'. The pilot was aware that the car carrier was to the north of its intended track but he perceived the action he was taking to head to the southern side of the buoyed channel was as substantial as the navigational constraints allowed. The pilot was under the impression that the vessel was heading to the south, whereas its heading was not altered significantly beyond the axis of the channel (**Figure 12**) until collision was imminent. Consequently, *City of Rotterdam* remained on the northern side of the buoyed channel and in the path of the inbound *Primula Seaways*.

The bridge teams on board both *City of Rotterdam* and *Primula Seaways* and the VTS operators were appropriately qualified and experienced. However, the interventions made by VTS were not sufficiently robust to make the pilot appreciate that more aggressive action was required to avoid the developing collision situation. Moreover, the absence of any challenge or intervention by *City of Rotterdam's* bridge team until collision was imminent indicates an over-reliance on the pilot and a breakdown in the bridge resource management on that vessel. In addition, although *Primula Seaways'* bridge team had identified the risk of collision and had taken action to clarify the pilot's intentions, substantial action to avoid the collision was taken too late to be effective.

2.3 RELATIVE MOTION ILLUSION

City of Rotterdam's pilot's spatial awareness was distorted during the 5 minutes preceding the vessel's collision with *Primula Seaways*. During this period, he had been drawn to the bridge window located directly above the starboard VHF radio that was angled off the perpendicular to the vessel's centreline. As a result, the pilot experienced a relative motion illusion in which the vessel appeared to be heading in the direction in which he was looking. The likelihood of the pilot experiencing this illusion has been confirmed through an analysis of the information recorded on the vessel's VDR, simulation (paragraph 1.12), the experiences of pilots who have worked on board either *City of Rotterdam* or *City of St Petersburg* (paragraph 1.10), and the experiences of MAIB inspectors and an ergonomist during pilotage transits in and out of Newcastle, UK (paragraph 1.11).

During the passage along the Bull Channel, *City of Rotterdam's* pilot had moved between the navigation workstation and the forward conning position as he monitored the vessel's position by eye, radar and ECS. The pilot had recognised that the car carrier had been set to the north of its intended track by the wind. He had also realised that he needed to manoeuvre the vessel further to the south in order

to pass *Primula Seaways* port to port. However, between the pilot calling the VTS at 2035 and the collision, it is likely that he would have spent much of the time looking through the window above the starboard VHF radio.

Relative motion illusion deceives a viewer into thinking that the observed view is the direction of travel. In this case, the pilot was looking through the window above the starboard VHF radio, which was 33° off-axis from the centreline. As it was dark, the inward slope of the window removed all objects in the periphery, and there were no visual clues, such as a forward structure or bow tip, the illusion would have been compelling.

Consequently, when during the VHF exchange with *Primula Seaways* at 2037 (**Table 3**) the pilot stated that “*I’m trying to drop her down to the south as much as possible is that you on my port bow?*” and “*I’m trying to bring her as far to the south as the wind will allow me but er [pause] yeah I’ll keep coming down to the south more positive*”, the pilot perceived that the vessel was moving towards the Bull anchorage, which was on a bearing of 138°, whereas the vessel was actually on a heading of 105°. In addition, *Primula Seaways* was on the starboard bow, but from the pilot’s perspective the ferry was on the left hand side of the window and would have appeared to have been on the car carrier’s port bow. That *Primula Seaways*’ second officer erroneously confirmed that the ferry was on the car carrier’s port bow would have reinforced the pilot’s illusion.

The pilot’s illusion remained during the VHF exchange with the VTS watch manager at 2038 (**Table 4**) when he stated “*we’re on full speed and heading nearly south but we’re going to try and bring her as far down as possible*” and “*I’m on full speed and heading right down south.*” At the time, the pilot was looking toward the Bull anchorage on a bearing of 148° whereas *City of Rotterdam* was heading 115° (**Figure 17**). The pilot had adjusted the vessel’s heading from 095° but he clearly was not cognisant of the difference between the heading he had steadied the ship, or a heading that he would typically have steered on passing 4A Clee Ness light float (**Figures 8 and 12**), and his perceived direction of travel.

As explained in the ergonomist’s report (**Annex B**), the pilot’s ability to reconcile the headings he had ordered with his perceived direction of travel was probably hindered by the fact that when navigating predominantly by eye, his egocentric frame of evidence would have been dominant. In short, he would have believed what he saw. It was also probably hindered by the cognitive cost when transferring between egocentric and exocentric frames of reference, and that a relative motion illusion is likely to return after being broken, even with regular reminders.

2.4 OVER-RELIANCE ON THE PILOT

Pilots trade on their reputation as ship handling experts with the local knowledge to ensure a vessel’s safe entry and departure from ports. Many masters rely heavily on them, defer to their knowledge and, in many cases, leave them to navigate the vessel without interference or supervision. The collision between the general cargo ship *Skagern* and the container vessel *Samskip Courier*, which resulted in MAIB recommendations (paragraph 1.15.5) is one of many similar accidents where ships’ crews’ over-reliance on pilots has been a contributing factor.

In this case, that neither *City of Rotterdam*’s master nor the third officer challenged the pilot’s actions until seconds before the collision was pivotal to the accident. The master and the pilot had discussed the departure plan before *City of Rotterdam*

sailed from Immingham. Both were aware of the likely effect of the wind on the high-sided vessel that was in ballast and the potential for disorientation when conning from a position away from the centreline or the navigation workstation. In such circumstances, the need for the bridge team to remain vigilant and to support, monitor and challenge the pilot's actions was compelling.

However, the master and the third officer left the responsibility for *City of Rotterdam*'s safe passage predominantly to the pilot soon after the vessel entered the main navigation channel. Although the third officer periodically plotted the vessel's position on the paper chart, **Figure 12** shows that these plots were inaccurate. Therefore, it is likely that the third officer's action was solely procedural, and that he was relying more on the ECS than the paper charts to monitor *City of Rotterdam*'s position. Nonetheless, neither he nor the master queried the pilot's course alteration to 095° at 2032 (**Figure 6**). Had the pilot's actions been closely monitored, it would have been apparent that, as 095° was the intended course to make good (**Figure 12**), the pilot had not taken into account the wind and tidal stream, which he had warned would have most effect in this area.

The master and the third officer also readily accepted the pilot's corrective actions to regain track, despite the challenges to *City of Rotterdam*'s position and movement by *Primula Seaways* at 2037 (**Table 3**) and by the VTS at 2038 (**Table 4**). That neither of these external challenges prompted a challenge from the master, even though *City of Rotterdam*'s position in the channel was contrary to the requirements of Rule 9 of the COLREGS, and the vessel was in the way of the inbound ferry, indicates that the bridge team had placed total reliance and trust on the pilot's ability to conn the vessel. Indeed, the master's comment at 2040:02 regarding *Primula Seaways* of "what's he doing?" suggests that the master had not kept abreast of the developing situation. Consequently, his direct intervention 14 seconds before the collision was far too late to be effective.

The ability of individuals to challenge decisions not forming part of an agreed plan is a fundamental aspect of bridge resource management, which is included in STCW training requirements. However, it is evident that the principles of such training, which also include the continuous monitoring of a vessel's progress to maintain safe passage, were not followed on this occasion. The applicable requirements of the onboard SMS manual (paragraph 1.6.2) with regard to the master's and third officer's responsibilities when the vessel was under pilotage, were also not met.

2.5 ACTION ON BOARD *PRIMULA SEAWAYS*

Primula Seaways' bridge team noticed that *City of Rotterdam* was on the northern side of the Bull Channel between the South Shoal buoy and 4A Clee Ness light float. The car carrier was on the 'wrong' side of the channel. Therefore, the master's query to the VTS at 2034 (**Table 1**) regarding the car carrier's intended movement was timely and appropriate. The vessels were about 3nm apart and the ferry's speed had been reduced to 16.5kts.

It is apparent from *Primula Seaways*' second officer's VHF call to *City of Rotterdam* at 2037 (**Table 3**) that the ferry's bridge team had continued to monitor *City of Rotterdam* and were concerned that the vessel was still on the northern side of the Bull Channel. However, the pilot assured the second officer that he was manoeuvring the car carrier to the south, an action that the pilot reiterated shortly

after being advised by the VTS that *City of Rotterdam* was heading into the Hawke anchorage. Nonetheless, *Primula Seaways*' master remained sufficiently concerned to set the ferry's engines to 'half ahead'.

The simulations of the accident (paragraph 1.11) showed that, when the ferry's speed was reduced at 2038:25, a course alteration to starboard would have caused the ferry to cross *City of Rotterdam*'s bow while an alteration to port would have exacerbated the situation if the pilot manoeuvred the car carrier as stated (**Figures 8 and 18**). Doubt clearly existed and, as the vessels were only 1nm apart and closing at a speed of 28.5kts, a more substantial reduction in speed was warranted. By the time full starboard helm was applied and the engine was set to 'full astern' at 2039:37, the vessels' closing speed was still 26kts and they were only 0.5nm apart (**Figures 19 and 20**).

2.6 VTS INTERVENTION

City of Rotterdam's progress in the Bull Channel was closely monitored by the VTS and the vessel's northerly position after it had rounded the South Shoal buoy was quickly detected. The VHF channel 12 operator communicated his concern to the watch manager and the situation continued to be monitored. In the circumstances, the watch manager's assessment that there was sufficient time to correct the situation was appropriate.

Nonetheless, an opportunity to raise concern at the vessel's position with the pilot and to highlight the approach of *Primula Seaways* when the pilot reported on passing 4A Clee Ness light float at 2035 (**Table 2**), was not taken. Although this was an opportunity missed, particularly as the call from the ferry to the VTS 1 minute earlier (**Table 1**) queried the route taken by *City of Rotterdam*, such a 'soft' challenge was unlikely to have broken the pilot's illusion. The illusion was not even broken by the VTS watch manager's intervention at 2038 (**Table 4**) when he advised the pilot that the vessel was entering the Hawke anchorage.

In hindsight, however, the watch manager's intervention could have been more effective had it been prefixed with a 'warning' message marker and had the watch manager not referred to the pilot by name. A 'warning' message to *City of Rotterdam* would have been more likely to have prompted the master to take more interest and challenge the pilot's actions. It would have also highlighted the VTS's concerns to *Primula Seaways*, which might have prompted an earlier and more substantial speed reduction on board the ferry. By the time the VTS warned *Primula Seaways* of the risk of collision, it was too late for further avoiding action to be taken.

2.7 CONSEQUENCES OF BRIDGE DESIGN

The aerodynamic shape of *City of Rotterdam*'s bow impacted on the design of the vessel's bridge, particularly with regard to the bridge windows. The inward sloping windows were contrary to the requirements of SOLAS V/22 1.9.1, but this was recognised by the shipbuilder. The precaution of moving sources of light away from the windows in order to reduce the likelihood of reflections, met the intent of the regulation, which enabled the PMA to issue exemptions.

However, the bridge shape, the mitigation measures taken to reduce reflections, and the equipment layout had consequences. In particular, although the centreline conning position had a 'commanding view', a close view of the sea surface (IACS

Recommendation 95) and it was possible to sight the azimuth repeater and the rudder angle indicator. However, the propulsion, wind and speed indicators on the deck head in front of the helmsman (**Figure 16**) were set back and were not readable as required by SOLAS V/19.2.5.4.

More significantly, with respect to the requirements of SOLAS IV/6 for radio installations, the VHF radios at the front of the bridge were “convenient” to the forward conning position, in that there was easy and unobstructed access to them. However, their location by the off-axis windows did not “ensure the greatest possible degree of safety” due to the potential for relative motion illusion. In this case, *City of Rotterdam*’s pilot’s relative motion illusion occurred as a result of him having to move away from the centreline window to use the starboard VHF radio.

The MAIB is not aware of relative motion illusion occurring on board other vessels with semi-circular shaped bridges, although these tend to be fitted with integrated navigation systems (paragraph 1.12). Therefore, it is understandable why the potential for the illusion was not foreseen and taken into account during the design of *City of St Petersburg* and *City of Rotterdam*. However, the need to learn from this accident and recognise the potential for the illusion to develop by off-axis windows in future designs is compelling. That *City of Rotterdam* and *City of St Petersburg* had operated for 5 years without navigational accidents, was largely due to the measures adopted on board the vessels such as the cord on the centreline window, the extended lead on the VHF radio (*City of St Petersburg*) and the coping strategies adopted by the ship’s crew and by the pilots (paragraph 1.9).

2.8 REGULATION OF INNOVATIVE DESIGNS

Innovations in ship and bridge design have the potential to make positive contributions to vessel safety and the environment. As such, they are to be encouraged, and it is inevitable that exemptions to particular criteria will be necessary to enable unconventional designs to enter into service. However, the circumstances of this case and the departures from good practice highlighted in the ergonomic assessment (paragraph 1.10 and **Annex B**) indicate that a stricter adherence to the ergonomic principles of bridge design detailed in SOLAS V/15 could reduce the likelihood of human error.

IACS Recommendation 95 (paragraph 1.14.2) provides important guidance on the implementation of international regulations and instruments related to bridge design. It also explains key terms such as ‘conning position’, which are referred to in SOLAS IV and SOLAS V but are not defined. However, like the guidelines in MSC.Circ. 982 (paragraph 1.14.1), the recommendations are not binding. Consequently, the ergonomic principles detailed in SOLAS V/15 are open to interpretation.

The IMO’s MSC decided in 2004 that the development of a UI that could be used to demonstrate compliance with SOLAS V/15 was not necessary. However, the freedom of interpretation and discretion this decision afforded to naval architects in the application of good practice increases the potential for innovative designs to have unintended consequences on human performance. Therefore, the need for an IACS UI on the interpretation of the ergonomic principles of bridge design warrants reconsideration.

SECTION 3 - CONCLUSIONS

3.1 SAFETY ISSUES DIRECTLY CONTRIBUTING TO THE ACCIDENT THAT HAVE BEEN ADDRESSED OR RESULTED IN RECOMMENDATIONS

1. The collision stemmed from *City of Rotterdam* being set to the northern side of the Bull Channel by the wind and the tidal stream followed by the distortion of its pilot's spatial awareness due to a 'relative motion illusion'. [2.2]
2. *City of Rotterdam's* pilot's relative motion illusion deceived him into thinking that his view from the window above the starboard VHF radio, which was 33° off the vessel's centreline axis, was the vessel's direction of travel. [2.3]
3. As it was dark, the inward slope of the window removed all objects in the pilot's periphery, and there were no visual clues such as a forward structure or bow tip, the illusion would have been compelling. [2.3]
4. The pilot's ability to reconcile the headings he had ordered with his perceived direction of travel was probably hindered by further psychological effects of the relative motion illusion, such as the cognitive costs of transferring between different frames of reference. [2.3]
5. Soon after *City of Rotterdam* entered the main navigation channel, the master and the third officer left the responsibility for the vessel's safe passage predominantly to the pilot. [2.4]
6. *City of Rotterdam's* master and third officer did not challenge the pilot's actions despite concern at the vessel's position being expressed by *Primula Seaways* and the VTS. The master's intervention, 14 seconds before the collision, was far too late to be effective. [2.4]
7. *City of Rotterdam's* bridge team's over reliance on the pilot, and its lack of effective monitoring of the vessel's progress, were evidence of ineffective bridge resource management. [2.4]
8. Although *Primula Seaways'* engines were reduced to 'half ahead' 2 minutes before the collision, a more substantial reduction of speed was warranted in view of the doubt concerning *City of Rotterdam's* movement. [2.5]
9. The VTS intervention at 2038 could have been more effective in alerting the bridge teams on board both vessels to its concerns had it been prefixed with a 'warning' message marker and it had not referred to the pilot by name. [2.6]
10. The location of the VHF radios by the off-axis windows on board *City of Rotterdam* increased the potential for relative motion illusion. [2.7]
11. The potential for relative motion illusion was unforeseen and therefore not taken into account during the design of *City of St Petersburg* and *City of Rotterdam*. [2.7]
12. Stricter adherence to the ergonomic principles of bridge design detailed in SOLAS V/15 would reduce the likelihood of human error. Therefore, the need for an IACS UI on the interpretation of the ergonomic principles of bridge design warrants reconsideration. [2.8]